Task 3

Journal:

*Tidbit - FC vs CNN

- How many weights does a 32x32 greyscale image have if the first FC layer has 512 neurons? 32x32x512 = 524,288 neurons
- How many weights does a 3x3 filter have on the same image? 3x3 = 9 neurons.
- Why does the CNN number stay the same no matter how large the image gets? The filter is reused across the entire image so the weight won't change.

- Kernel vs Filter: A kernel is the small 2D weight matrix applied to 1 input channel. A filter is the full 3D structure, kernel x input channels.

* Tidbit - Compute One output cell

Input

7	2	9	0	4
6	0	2	7	0
1	7	6	5	9
8	2	8	3	1
3	5	9	6	4

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Kernel

2	0	-2
-1	1	0
0	-1	-1

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Output

-23	-5	1
4	-26	-9
-30	11	-21

$$[1,1] = \begin{matrix} 14+0+-18 \\ -6+0+0 \\ 0+(-7)+(-6) \end{matrix} = -23 \qquad [1,2] = \begin{matrix} 4+0+0 \\ 0+2+0 \\ 0+(-6)+(-5) \end{matrix} = -5 \qquad [1,3] = \begin{matrix} 18+0+(-8) \\ -2+7+0 \\ 0+(-5)+(-9) \end{matrix} = 1$$

$$[2,1] = \begin{matrix} 12+0+(-4) \\ -1+7+0 \\ 0+(-2)+(-8) \end{matrix} = 4 \qquad [2,2] = \begin{matrix} 0+0+(-14) \\ -7+6+0 \\ 0+(-8)+(-3) \end{matrix} = -26 \qquad [2,3] = \begin{matrix} 4+0+0 \\ -6+5+0 \\ 0+(-3)+(-9) \end{matrix} = -9$$

$$[3,1] = \begin{matrix} 2+0+(-12) \\ -8+2+0 \\ 0+(-5)+(-9) \end{matrix} = -30 \qquad [3,2] = \begin{matrix} 14+6+0 \\ -2+8+0 \\ 0+(-9)+(-6) \end{matrix} = 11 \qquad [3,3] = \begin{matrix} 12+0+(-18) \\ -8+3+0 \\ 0+(-6)+(-4) \end{matrix} = -21$$

* Tidbit - Edge Detectors (same 5x5 matrix)

Vertical Edge Detector

Input

7	2	9	0	4
6	0	2	7	0
1	7	6	5	9
8	2	8	3	1
3	5	9	6	4

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Detector

-1	0	1
-2	0	2
-1	0	1

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Output

1	10	-6
6	4	-4
11	2	-15

$$[1,1] = \begin{matrix} -7+0+9 \\ -12+0+4 \\ -1+0+6 \end{matrix} = 1 \qquad [1,2] = \begin{matrix} -2+0+0 \\ 0+0+14 \\ -7+0+5 \end{matrix} = 10 \qquad [1,3] = \begin{matrix} -9+0+4 \\ -4+0+0 \\ -6+0+9 \end{matrix} = -6$$

$$[2,1] = \begin{matrix} -6+0+2 \\ -2+0+12 \\ -8+0+8 \end{matrix} = 6 \qquad [2,2] = \begin{matrix} 0+0+7 \\ -14+0+10 \\ -2+0+3 \end{matrix} = 4 \qquad [2,3] = \begin{matrix} -2+0+0 \\ -12+0+18 \\ -8+0+1 \end{matrix} = -4$$

$$[3,1] = \begin{matrix} -1+0+6 \\ -16+0+16 \\ -3+0+9 \end{matrix} = 11 \qquad [3,2] = \begin{matrix} -7+0+5 \\ -4+0+6 \\ -5+0+6 \end{matrix} = 1 \qquad [3,3] = \begin{matrix} -6+0+9 \\ -16+0+2 \\ -9+0+4 \end{matrix} = -15$$

Horizontal Edge Detector

Input

7	2	9	0	4
6	0	2	7	0
1	7	6	5	9
8	2	8	3	1
3	5	9	6	4

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Detector

-1	-2	-1
0	0	0
1	2	1

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Output

1	4	12
12	10	-1
1	5	0

$$[1,1] = \begin{matrix} -(7+4+9) \\ 0+0+0 \\ 1+14+6 \end{matrix} = 1 \qquad [1,2] = \begin{matrix} -(2+18+0) \\ 0+0+0 \\ 7+12+5 \end{matrix} = 4 \qquad [1,3] = \begin{matrix} -(9+0+4) \\ 0+0+0 \\ 6+10+9 \end{matrix} = 12$$

$$[2,1] = \begin{matrix} -(6+0+2) \\ 0+0+0 \\ 8+4+8 \end{matrix} = 12 \qquad [2,2] = \begin{matrix} -(0+4+7) \\ 0+0+0 \\ 2+16+3 \end{matrix} = 10 \qquad [2,3] = \begin{matrix} -(2+14+0) \\ 0+0+0 \\ 8+6+1 \end{matrix} = -1$$

$$[3,1] = \begin{matrix} -(1+14+6) \\ 0+0+0 \\ 3+10+9 \end{matrix} = 1 \qquad [3,2] = \begin{matrix} -(-7+12+5) \\ 0+0+0 \\ 5+18+6 \end{matrix} = 5 \qquad [3,3] = \begin{matrix} -(6+10+9) \\ 0+0+0 \\ 9+12+4 \end{matrix} = 0$$

- The closer each output value is to 0, the less likely it is to be an edge.

- Output size formula:

$$\text{Output size} = \frac{W-K+2P}{S}+1$$

W = input spatial size
K = kernel size
P = padding
S = Stride

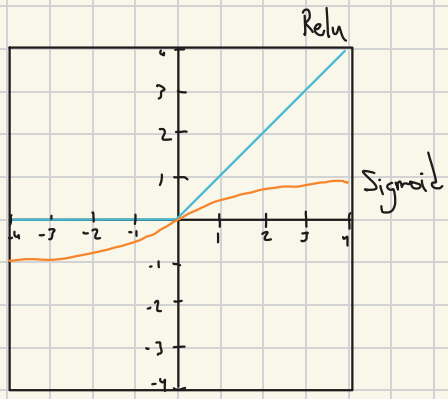
* Tidbit - Output size calculations

- W=64, K=5, P=0, S=1 : 60
- W=64, K=3, P=1, S=1 : 64
- W=64, K=3, P=0, S=2 : 31
- What padding keeps W=128 unchanged with K=5, S=1? $P = \frac{K-1}{2} = \frac{4}{2} = 2$

* Tidbit - Channels & Parameters

- How many learning weights does a Conv layer with 32 filters, kernel size 3x3 and 3 input channels have? 3x3 x 3 x 32 = 864
- How many weights will it have if the next layer has 64 filters with 3x3 kernels? 3x3x32x64
- Why does adding more filters increase depth but not spatial size? Increasing the size would mean adding information that isn't in the input. They can only decrease or maintain the same size.

* Tidbit - Activation Function



* Tidbit - Pooling

7	2	9	0
6	0	2	7
1	7	6	5
8	2	8	3

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7	9
8	8

Max

7	2	9	0
6	0	2	7
1	7	6	5
8	2	8	3

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3	4
4	5

Average

* Tidbit - Loss and Optimizers

- Why does Cross-Entropy penalise a confident wrong prediction more than an uncertain one? Because being almost right is better than being very wrong.

* Tidbit - Regularisation

- You randomly zero out 50% of neurons during training but not at test time. What technique is this? This is an example of the Dropout, its purpose is to improve generalization by forcing redundant representations of neurons.
- Your model gets 99% train accuracy and 62% validation accuracy. Name the problem and list two fixes. This is an example of overfitting. One fix could be to utilize the dropout technique to generalize the data, or you could increase the dataset size artificially or traditionally.